

University of Veterinary and Animal Sciences, Lahore

B.S (Hons) Food Science and Technology - Mathematics-II Mid Term Exam 2022

Complete Solutions with Theory

Question 1: Squeeze Theorem and Limits

Question:

State Squeeze theorem and find $\lim_{x \rightarrow 1} (x^2 + x - 2) \cdot \cos\left(\frac{1}{x-1}\right)$ by using squeeze theorem.

Theory:

Squeeze Theorem (Sandwich Theorem): If $f(x) \leq g(x) \leq h(x)$ for all x in some interval containing a (except possibly at a), and if:

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

Then:

$$\lim_{x \rightarrow a} g(x) = L$$

In other words, if a function is “squeezed” between two other functions that have the same limit at a point, then the squeezed function must also have that same limit.

Solution:

Step 1: Recognize that $-1 \leq \cos\left(\frac{1}{x-1}\right) \leq 1$ for all $x \neq 1$

Step 2: Multiply the inequality by $(x^2 + x - 2)$

Since $(x^2 + x - 2) = (x + 2)(x - 1)$, as $x \rightarrow 1$, this expression approaches $(1 + 2)(1 - 1) = 0$

We need to consider the sign: as $x \rightarrow 1$, $(x^2 + x - 2) \rightarrow 0$

Step 3: Set up the squeeze:

$$-(x^2 + x - 2) \leq (x^2 + x - 2) \cos\left(\frac{1}{x-1}\right) \leq (x^2 + x - 2)$$

Step 4: Find the limits of the bounding functions:

$$\lim_{x \rightarrow 1} -(x^2 + x - 2) = -(1 + 1 - 2) = 0$$

$$\lim_{x \rightarrow 1} (x^2 + x - 2) = 1 + 1 - 2 = 0$$

Step 5: Apply the Squeeze Theorem:

Since both bounding functions approach 0, by the Squeeze Theorem:

$$\lim_{x \rightarrow 1} (x^2 + x - 2) \cdot \cos\left(\frac{1}{x-1}\right) = 0$$

Question 2: Piecewise Function Domain

Question:

If $f(x) = \begin{cases} x & ; 0 \leq x \leq 1 \\ x - 1 & ; 1 \leq x \leq 2 \end{cases}$, then the domain of $f(x)$ is _____

Theory:

Domain of a Function: The domain of a function is the set of all possible input values (x-values) for which the function is defined. For piecewise functions, the domain is the union of all intervals where the function is defined.

Solution:

The function is defined in two pieces: - First piece: x is defined for $0 \leq x \leq 1$ - Second piece: $x - 1$ is defined for $1 \leq x \leq 2$

The domain is the union of these intervals:

$$\text{Domain} = [0, 1] \cup [1, 2] = [0, 2]$$

Answer: $[0, 2]$

Question 3: Range of a Function

Question:

If $f(x) = \sqrt{16 - x^4}$, then the range of $f(x)$ is: - a. $[0, 4]$ - b. $(0, 4)$ - c. $[0, 2]$ - d. $\mathbb{R}$

Theory:

Range of a Function: The range is the set of all possible output values (y-values) that a function can produce. For square root functions, we need to: 1. Find the domain (values under the square root must be non-negative) 2. Determine the minimum and maximum values of the function

Solution:

Step 1: Find the domain For $f(x)$ to be defined: $16 - x^4 \geq 0$

$$x^4 \leq 16$$

$$|x| \leq 2$$

$$-2 \leq x \leq 2$$

Step 2: Find the range - The expression inside the square root: $16 - x^4$ - Maximum value occurs when $x = 0$: $16 - 0 = 16$ - So $f(0) = \sqrt{16} = 4$ - Minimum value occurs when $x = \pm 2$: $16 - 16 = 0$ - So $f(\pm 2) = \sqrt{0} = 0$

Step 3: Since $f(x) = \sqrt{16 - x^4}$ is continuous on $[-2, 2]$ and takes all values from 0 to 4:

Answer: a. $[0, 4]$

Question 4: Domain of Fraction with Variable

Question:

The domain of $\frac{x}{y}$ is _____

Theory:

Domain of Rational Functions: For a function $\frac{f(x)}{g(x)}$, the domain consists of all values where: 1. Both $f(x)$ and $g(x)$ are defined 2. $g(x) \neq 0$ (denominator cannot be zero)

For multivariable expressions, we consider all variables.

Solution:

For $\frac{x}{y}$ to be defined: - x can be any real number - y must be non-zero (division by zero is undefined)

Answer: $\{(x, y) : x \in \mathbb{R}, y \in \mathbb{R} \setminus \{0\}\}$

Or simply: $y \neq 0, x \in \mathbb{R}$

Question 5: Composition of Functions

Question:

Express the function $F(x) = \frac{1}{\sqrt{x+\sqrt{x}}}$ as a composition of three functions.

Theory:

Function Composition: A composition of functions means applying one function to the result of another. If $F = f \circ g \circ h$, then:

$$F(x) = f(g(h(x)))$$

To decompose a complex function, we work from the inside out, identifying nested operations.

Solution:

Let's identify the three functions by working from inside to outside:

Step 1: Innermost function - $h(x) = \sqrt{x}$

Step 2: Middle function - $g(u) = x + u$ where $u = h(x)$ - More specifically: $g(x) = x + \sqrt{x}$

Step 3: Outer function - $f(v) = \frac{1}{\sqrt{v}}$ where $v = g(x)$

Verification: - $h(x) = \sqrt{x}$ - $g(h(x)) = x + \sqrt{x}$ - $f(g(h(x))) = \frac{1}{\sqrt{x+\sqrt{x}}} = F(x)$

Alternative decomposition:

Let: - $h(x) = x + \sqrt{x}$ - $g(u) = \sqrt{u}$ - $f(v) = \frac{1}{v}$

Then: $F(x) = f(g(h(x))) = \frac{1}{\sqrt{x+\sqrt{x}}}$

Answer: $f(x) = \frac{1}{x}, \quad g(x) = \sqrt{x}, \quad h(x) = x + \sqrt{x}$

Or: $F = f \circ g \circ h$ where $f(x) = \frac{1}{x}, g(x) = \sqrt{x}, h(x) = x + \sqrt{x}$

Question 6: Circle Equation - Top Half and Domain

Question:

Evaluate the top half of the circle $x^2 + (y - 2)^2 = 4$. Also find its domain.

Theory:

Circle Equation: The standard form of a circle is $(x - h)^2 + (y - k)^2 = r^2$ where: - Center: (h, k) - Radius: r

To find the top half, solve for y and take the positive square root. The domain consists of all x -values for which the circle exists.

Solution:

Given equation: $x^2 + (y - 2)^2 = 4$

Step 1: Identify circle properties - Center: $(0, 2)$ - Radius: $r = 2$

Step 2: Solve for y :

$$\begin{aligned}(y - 2)^2 &= 4 - x^2 \\ y - 2 &= \pm\sqrt{4 - x^2} \\ y &= 2 \pm \sqrt{4 - x^2}\end{aligned}$$

Step 3: Top half (take positive square root):

$$y = 2 + \sqrt{4 - x^2}$$

Step 4: Find the domain: For the square root to be defined: $4 - x^2 \geq 0$

$$\begin{aligned}x^2 &\leq 4 \\ |x| &\leq 2 \\ -2 &\leq x \leq 2\end{aligned}$$

Answer: - Top half: $y = 2 + \sqrt{4 - x^2}$ - Domain: $[-2, 2]$

Question 7: Limit Evaluation

Question:

$\lim_{x \rightarrow 2} \frac{x-1}{x}$ is _____

Theory:

Direct Substitution Method: If a function is continuous at a point a , then:

$$\lim_{x \rightarrow a} f(x) = f(a)$$

This is the simplest method for evaluating limits when the function is defined and continuous at the point.

Solution:

The function $f(x) = \frac{x-1}{x}$ is continuous at $x = 2$ (no division by zero or undefined operations).

Direct substitution:

$$\lim_{x \rightarrow 2} \frac{x-1}{x} = \frac{2-1}{2} = \frac{1}{2}$$

Answer: $\boxed{\frac{1}{2}}$

Question 8: Absolute Value Function**Question:**

Define absolute value function with example.

Theory:

Absolute Value Function: The absolute value function, denoted as $|x|$ or $f(x) = |x|$, gives the distance of a number from zero on the number line, regardless of direction. It is always non-negative.

Definition:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Properties:

1. $|x| \geq 0$ for all $x \in \mathbb{R}$
2. $|x| = 0$ if and only if $x = 0$
3. $|-x| = |x|$ (even function)
4. $|xy| = |x||y|$
5. $|x + y| \leq |x| + |y|$ (Triangle inequality)

Examples:

Example 1: Basic values - $|5| = 5$ (positive number stays positive) - $|-5| = 5$ (negative number becomes positive) - $|0| = 0$ (zero is zero)

Example 2: Solve $|x - 3| = 5$

$$\begin{aligned} x - 3 = 5 & \quad \text{or} \quad x - 3 = -5 \\ x = 8 & \quad \text{or} \quad x = -2 \end{aligned}$$

Example 3: Graph of $f(x) = |x|$ - V-shaped graph with vertex at origin - Slope is -1 for $x < 0$ and +1 for $x > 0$

Example 4: Practical application - Distance: $|x_2 - x_1|$ gives distance between points - If you're at position 3 and move to position 8: distance = $|8 - 3| = 5$ - If you're at position 8 and move to position 3: distance = $|3 - 8| = |-5| = 5$

Question 9: Composite Functions and Domain

Question:

If $f(x) = \sqrt{x+1}$ and $g(x) = \frac{1}{x^2-1}$; find $(g \circ f)(x)$. Also find its domain.

Theory:

Composite Function: $(g \circ f)(x) = g(f(x))$ means we substitute $f(x)$ into $g(x)$.

Domain of Composite Function: For $(g \circ f)(x)$ to be defined: 1. x must be in the domain of f 2. $f(x)$ must be in the domain of g

Solution:

Step 1: Find $(g \circ f)(x)$

$$(g \circ f)(x) = g(f(x)) = g(\sqrt{x+1})$$

Substitute into $g(x)$:

$$g(\sqrt{x+1}) = \frac{1}{(\sqrt{x+1})^2 - 1} = \frac{1}{x+1-1} = \frac{1}{x}$$

$$(g \circ f)(x) = \frac{1}{x}$$

Step 2: Find the domain

Conditions: 1. For $f(x) = \sqrt{x+1}$: $x+1 \geq 0 \Rightarrow x \geq -1$ 2. For $g(f(x)) = \frac{1}{x}$: $x \neq 0$ 3. Also, $f(x)$ must be in domain of g : $(\sqrt{x+1})^2 - 1 \neq 0 \Rightarrow x \neq 0$

Combining all conditions:

$$x \geq -1 \quad \text{AND} \quad x \neq 0$$

Domain: $[-1, 0) \cup (0, \infty)$ or $[-1, \infty) \setminus \{0\}$

Question 10: Range and Domain of Cosine Function

Question:

If $f(x) = \cos x$, the range of f is $[-1, 1]$, then the domain of $f(x)$ is _____

Theory:

Cosine Function: - The cosine function is defined for all real numbers - It is a periodic function with period 2π - Range: $[-1, 1]$ (bounded between -1 and 1) - Domain: All real numbers (no restrictions)

Solution:

The cosine function $f(x) = \cos x$ is defined for all real numbers. There are no restrictions such as: - Division by zero - Square roots of negative numbers - Logarithms of non-positive numbers

Answer: $\mathbb{R}$ or $(-\infty, \infty)$

The domain is all real numbers.

Question 11: Domain and Range of Absolute Value

Question:

The domain and range of $f(x) = |x|$ is: - a. $(-\infty, 0); (0, \infty)$ - b. $(-\infty, \infty); (-\infty, 0]$ - c. $(-\infty, \infty); [0, \infty)$ - d. None of these

Theory:

Absolute Value Function $f(x) = |x|$: - **Domain:** The function accepts any real number as input - **Range:** The output is always non-negative (zero or positive)

Solution:

Domain Analysis: $|x|$ is defined for all real numbers. There are no restrictions.

$$\text{Domain} = (-\infty, \infty) = \mathbb{R}$$

Range Analysis: - For $x \geq 0$: $|x| = x \geq 0$ - For $x < 0$: $|x| = -x > 0$ - Minimum value: $|0| = 0$ - The function can take any non-negative value

$$\text{Range} = [0, \infty)$$

Answer: c. $(-\infty, \infty); [0, \infty)$

Question 12: Symmetry of Even Functions

Question:

The graph of an Even function is symmetric about the _____

Theory:

Even Functions: A function $f(x)$ is even if and only if:

$$f(-x) = f(x) \quad \text{for all } x \text{ in the domain}$$

Symmetry: Even functions are symmetric about the **y-axis**. This means if you fold the graph along the y-axis, the two halves would match perfectly.

Examples of Even Functions: - $f(x) = x^2$ (parabola) - $f(x) = |x|$ (V-shape) - $f(x) = \cos x$ - $f(x) = x^4 + x^2 + 1$

Odd Functions (for comparison): - Symmetric about the origin - $f(-x) = -f(x)$ - Examples: $f(x) = x$, $f(x) = x^3$, $f(x) = \sin x$

Solution:

Answer: y-axis

The graph of an even function is symmetric about the **y-axis**.

Question 13: Epsilon-Delta Definition of Limit

Question:

Define $\epsilon - \delta$ limits of a function.

Theory:

Precise Definition of a Limit (Epsilon-Delta Definition):

We say that:

$$\lim_{x \rightarrow a} f(x) = L$$

if and only if:

For every $\epsilon > 0$, there exists a $\delta > 0$ such that:

$$\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - L| < \epsilon$$

Interpretation:

- ϵ (**epsilon**): Represents how close we want $f(x)$ to be to the limit L . It's the "tolerance" or "error margin" for the output.
- δ (**delta**): Represents how close x needs to be to a to achieve the desired closeness of $f(x)$ to L . It's the "tolerance" for the input.
- $0 < |x - a|$: This ensures $x \neq a$ (we don't care about the function value at a itself, only nearby values).
- $|x - a| < \delta$: x is within a distance δ from a .
- $|f(x) - L| < \epsilon$: $f(x)$ is within a distance ϵ from L .

Informal Explanation:

The limit of $f(x)$ as x approaches a is L if we can make $f(x)$ arbitrarily close to L by taking x sufficiently close to a (but not equal to a).

Example:

Prove that $\lim_{x \rightarrow 2} (3x + 1) = 7$ using the ϵ - δ definition.

Proof: Let $\epsilon > 0$ be given. We need to find $\delta > 0$ such that:

$$\text{if } 0 < |x - 2| < \delta, \text{ then } |(3x + 1) - 7| < \epsilon$$

Simplify:

$$|(3x + 1) - 7| = |3x - 6| = |3(x - 2)| = 3|x - 2|$$

We want: $3|x - 2| < \epsilon$

Therefore: $|x - 2| < \frac{\epsilon}{3}$

Choose $\delta = \frac{\epsilon}{3}$

Then, if $0 < |x - 2| < \delta = \frac{\epsilon}{3}$:

$$|(3x + 1) - 7| = 3|x - 2| < 3 \cdot \frac{\epsilon}{3} = \epsilon$$

This proves the limit.

Additional Questions from Other Papers:

Question A1: Multiple Composite Functions

Question:

- (a) If $g(x) = \sqrt{x} - 4$ and $h(x) = x^2 - 4$, find $(h \circ g)(x)$. Also find its domain.
(b) Find the domain of the function $f(x) = \frac{1}{x^2 - 3x}$

Solution A1(a):

Step 1: Find $(h \circ g)(x)$

$$(h \circ g)(x) = h(g(x)) = h(\sqrt{x} - 4)$$

Substitute into $h(x) = x^2 - 4$:

$$\begin{aligned} h(\sqrt{x} - 4) &= (\sqrt{x} - 4)^2 - 4 \\ &= x - 8\sqrt{x} + 16 - 4 \\ &= x - 8\sqrt{x} + 12 \end{aligned}$$

$$(h \circ g)(x) = x - 8\sqrt{x} + 12$$

Step 2: Find domain For $g(x) = \sqrt{x} - 4$ to be defined: $x \geq 0$

No additional restrictions from $h(g(x))$.

Domain: $[0, \infty)$

Solution A1(c):

For $f(x) = \frac{1}{x^2 - 3x}$ to be defined:

$$\begin{aligned} x^2 - 3x &\neq 0 \\ x(x - 3) &\neq 0 \\ x &\neq 0 \text{ and } x \neq 3 \end{aligned}$$

Domain: $\mathbb{R} \setminus \{0, 3\}$ or $(-\infty, 0) \cup (0, 3) \cup (3, \infty)$
